

PLFD : Statistical Analysis Plan

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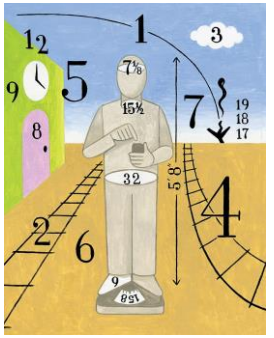
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Overview

- What is a Statistical Analysis Plan?
- Why do we need one?
- Sections in a Statistical Analysis Plan?

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What is a SAP ?



- SAP = Statistical Analysis Plan
- Also called Data Analysis Plan

- Plan = something you do "prior to"

- Technical and detailed document outlining the planned statistical analysis of the study objectives

- Roadmap of data and proposed statistical analysis. So what data do you have, what are you going to do with it and for what reason

What is a SAP ?

- So, basically it describes what variables and outcomes will be collected and which statistical methods will be used to analyze them.
  - Describes how the data is handled and processed
  - Guides the data analysis from the preliminary stage of summarizing and describing the data through testing of the hypotheses and answer the objectives of the study
  - Describes when conclusions are made and how
  - Specifies the software (and version) that is used

Why do we need a SAP ?

- Why do you as a scientist need a SAP ?
- It is a tool to operationalize the objectives of your study in a clear way
  - You reflect on the data you will collect
    - Ensures you collect all data (variables) needed
    - Ensures all collected data is used
  - You think about when and how you will analyze your data
- You have a plan to follow, you can not just do what you want with the data so this will limit the risk that you start to torture your data, --- > ***reproducible results***

Why do we need a SAP ?

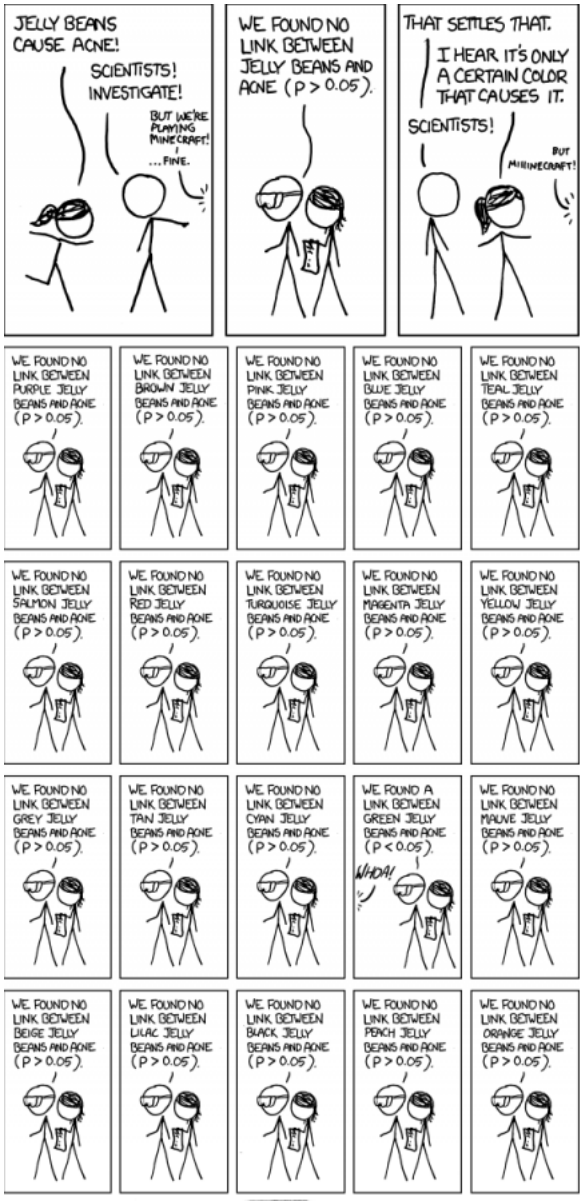
- Credibility of the research depends on a critical assessment by others of the strength and weakness in study design, conduct and analysis

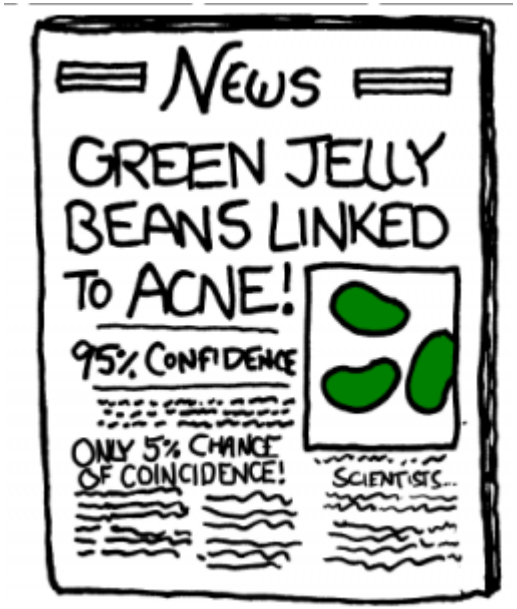
  → transparent reporting is needed (CONSORT, STROBE)

- To guarantee the reproducibility of study results by individuals not involved in the study, transparency is therefore essential
  - During the statistical analysis a multitude of decisions are taken and these have an impact on results and conclusions

- Minimizing selective outcome reporting, reporting bias

- To avoid p-hacking
  - Manipulating data and analysis until you get a statistically significant result





So, uh, we did the green study again and got no link



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How to start writing a SAP ?

- You need to have a clear **Research Question**

Frameworks to evaluate the quality of a RQ:

PICO (interventional study)

PECO (prospective cohort study)

P= population of interest

I= intervention , E = exposure

C = comparator

O= outcome

Research Question: In adults with chronic lower back pain, does yoga reduce pain intensity compared to standard physical therapy over 12 weeks?

P (Population): Adults with chronic lower back pain
I (Intervention): Yoga
C (Comparison): Standard physical therapy
O (Outcome): Reduction in pain intensity over 12 weeks

- Research Question informs the following:
  - Primary objectives
    - What are the outcomes ? What is the aim: prevalence estimation, effect of an intervention, studying associations ?
  - Secondary objectives
  - Methods (study design)
    - E.g. if objective is the estimation of a prevalence a RCT is not good, better cross-sectional study
  - Sample size

When these are set, a comprehensive SAP can be created

Steps

Aims of study

Design/Relevant Variables

- When and how measured
- Variable table
  - Nature of variable
  - Role of the variable

Appropriate Statistical Tests

- Depends on design and type of outcome
- Only descriptive stats or also inferential stats ?

Presentation of Results

- How will results be reported
- Mock tables/graphs

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When to write the SAP ?

- Ideally, when study is being designed
- Before examination of the data, before unblinding of data to the statistician
- Changes can be made to a SAP
  - Detail the changes, when were they made and for what reason
  - Include a version number/date in the SAP

→ Prevent 'data-dredging'

- pre-planned analysis <--> data driven exercises. Clearly describe which analysis (e.g. extra subgroup analyses) were done after looking at the data
- Idem for the construction of new variables (e.g. dichotomizing continuous variables)

Who writes the SAP ?

- Written by the study investigators and statistician
- Investigators:
  - Define study objectives
  - Study variables (outcomes, covariates, exposures)
- Statistician
  - Clarify research question or hypotheses
  - Identify outcome variables related to research question
  - Address issues of bias (attrition, confounding, …)
  - Sample size selection
  - Decide appropriate statistical methods

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SAP for RCT

- In the past, SAP was predominantly used in clinical trials
- Course on clinical trials for more info on clinical trial protocol and SAP

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SAP for Observational Study

- Observational studies complement RCT
  - RCT are costly and it takes time before results are available
  - Observational studies
    - provide information on safety (rare or late adverse events)
    - Study effectiveness in population
    - Research into causes of disease
- But the findings of observational studies are sometimes not replicable: not on the same data nor on new data
- Possible reasons for this:
  - Exploratory nature of observational questions
  - (Treatment) selection bias
  - Publication bias
  - Inadequate reporting (STROBE)
  - Tendency to pursue post hoc hypotheses (not prespecified hypothesis) = 'data dredging'

→ importance of developing a plan of analysis (SAP), before examination of the data

Sections in the SAP ?

- Administrative information
  - Title
  - Study registration number: ClinicalTrials.gov
  - Version and revision history, date
  - Names, affiliations, signatures

- Introduction
  - Background and rationale : research question and justification for undertaking the trial
  - Objectives and hypothesis
    - primary
    - secondary
  - E.g.
    - Null hypothesis: Students who eat breakfast will not perform better on a math exam than students who do not eat breakfast
    - Alternative hypothesis : Students who eat breakfast will perform better on a math exam than students who do not eat breakfast

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Sections in the SAP ?

- Study methods
  - Study design (e.g. case-control, cross-sectional, cohort, parallel group, ….)
  - Sample size calculations (at least) for primary analysis
- Study population
  - Eligibility criteria
  - Info on recruitment (flow diagram)
  - Withdrawal/follow-up/dropouts: timing, reasons,…

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Sections in the SAP ?

- Try to foresee decisions that are needed during the analysis and prespecify the choice in the SAP

- General Statistical Principles
  - Level of significance (statistical significance, clinical significance)
  - Type of hypothesis
    - Difference (one sided or two sided)
    - Equivalence
    - Non-inferiority
  - Confidence intervals to be reported
  - Adjustment for multiplicity: if so, what risk is controlled (FWER/FDR), at what nominal level by which adjustment method
    - Multiple primary endpoints ?
    - Multiple groups to be compared ?
    - Interim analysis ?
  - Definition of protocol deviations for the study
  - Definition of analysis population: e.g. per protocol, intention-to-treat, complete-case,…

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Sections in the SAP ?

- Description of all variables of interest
  - Definition, units, how and when are the outcomes measured
  - Nature of the variables: continuous, ordinal, time-to-event,…
  - Coding of missing values
  - Collapsing categories
  - Calculations, transformation, normalization of variables (e.g. change versus baseline, logarithmic transformation, …)
  - Specify role of variable in the analysis: outcome, predictor, confounding, mediator

  - E.g. Mental well-being is assessed by the Generalized Anxiety Disorder questionnaire (GAD – ref) at weeks 3, 6 and 12. The scores for Generalized Anxiety Disorder will be grouped according to:
    - Score 0-4: Minimal Anxiety
    - Score 5-9: Mild Anxiety
    - Score 10-14: Moderate Anxiety
    - Score greater than 15: Severe Anxiety
  - E.g. Cardiac output (L/min) = heart rate x VTI x 3.14 $(0.5 \times LVOT)^2$

Sections in the SAP

- Describe in detail the statistical techniques and presentation of results
  - the initial stage of summarizing and describing the data
  - testing the specified hypotheses (inferential statistics)



Sections in the SAP

Descriptive statistics (univariate analyses)

Eyeball your data - Summarizing and describing the data

- Goal: learn about the study sample (patient characteristics)
- Numeric description of the data - summary statistics

- Continuous variable
  - Range of values
  - Mean/median
  - Standard deviation / interquartile range

- Nominal/ordinal variable
  - Counts
  - %
  - Mode

- Typos or unrealistic values ?

Sections in the SAP

Descriptive statistics

Eyeball your data - Summarizing and describing the data

- Numeric description of the data - summary statistics
- Visualizations
  - Continuous variable
    - Histogram (normality ?needed ?)
    - Boxplot
    - Scatterplot for associations
  - Nominal/ordinal variable
    - Pie chart
    - Bar chart

Sections in the SAP

Descriptive statistics

- Results for sample in total, or per subgroup (e.g. m/f) or treatment/exposure group

- Information on the amount of missing information:
  - Report % of missing observations in each variable
  - How will you look for patterns of missing data (e.g. by gender),…
  - Comment on possible consequences

Statistical Plan

Inferential Statistics /modelling

- Goals: draw conclusions for the specified hypotheses, generalizable beyond study sample
- Motivate the appropriate statistical model for each endpoint
  - E.g. linear regression model, Cox-proportional hazard model, linear mixed model, ANOVA, t-test, …
- Depends on
  - Nature of the outcome variable : continuous, binary, count, time-to-event
  - Type of research question
    - Study correlation
    - Study association
    - Comparing groups
    - ….
  - Bivariate (unadjusted results) or multivariable analyses (adjusted results)
  - Study design
    - Cross sectional versus longitudinal data
    - Unpaired versus paired groups
    - Survey sample
    - …

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Statistical Plan

Inferential Statistics /modelling

- Describe the statistical model
  - Dependent variable/outcome
  - Explanatory variables:
    - Role (e.g. confounder, mediator,…)
    - Functional form for continuous variables (linear, quadratic)
    - Grouping of levels for categorical variables (if needed)
    - Interaction terms
    - Potential confounding variables, how dealt with in analysis
    - Correlated variables: multicollinearity
  - (Random effects)
  - Detail the process of (statistical) selection of explanatory variables
    - Variable selection procedure :e.g. forward, backward ,…
    - Specify the criteria involved
    - Significance levels

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Statistical Plan

Inferential Statistics /modelling

- Discuss what underlying assumptions are and how they will be checked, possible consequences, e.g.
  - Formal test for normality, linearity, goodness-of-fit statistics, …
  - Diagnostic plots: residual plots, QQ-plots, …

- Multicollinearity: how investigated, remedial actions

- Outliers: how investigated and dealt with

- Discuss solutions (remedial measures) in case of violations of the model assumptions
  - Transformations of the outcome variable, effect on interpretation of the results
  - Nonparametric alternative

Statistical Plan

Inferential Statistics /modelling

- Detail the procedure to deal with missing data
  - Complete case analysis
  - Imputation methods
  - …
- Subgroup analyses
  - Definition of the subgroup – how are they created based on the variables in the study
  - Purpose of subgroup analyses
    - Descriptive
    - Inferential
      - How ? Per subgroup or model with an interaction terms
      - Correction for multiple testing: FWER,FDR,…

Statistical Plan

- Analysis plan table

Objective	Outcome/ variable	Variable Type	Predictors/co mparisons	Analysis
Primary				
Primary				
….				
Secondary				

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Statistical Plan

- Analysis plan table

Objective	Outcome/ variable	Variable Type	Predictors/comparisons	Analysis
Determine difference in tumor size between the standard therapy and experimental therapy	Tumor size (mm^2)	Continuous	Treatment group (standard, experimental)	Scatter plot, linear regression

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Statistical Plan

Reporting

- Presentation of results
  - What are you going to present
    - E.g. : Means ? Proportions ? OR? Effect sizes ? Adjusted results ?
  - Parameter estimates and confidence intervals
    - Related to the research question
    - Related to covariates, confounders,.. ?
  - Mock tables, graphs
    - Typically
      - Table 1: Demographics, stratified by treatment group ?
      - Table 2: Primary Outcomes
      - Table 3: Secondary outcomes
- Software (version) for data collection/management and statistical analysis
- References

Statistical Plan

Reporting

- E.g. Mock Table 1 – Demographics

Variable	Treatment group		
	Control (n= xxx)	Experimental (n= xxx)	Total (n= xxx)
Age (years), mean (std)			
BMI (kg/m2), mean (sd)			
Gender (female), n (5)			
Smoking status never smoker, n (%) ex-smore, n(%) current-smoker, n (%)			

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